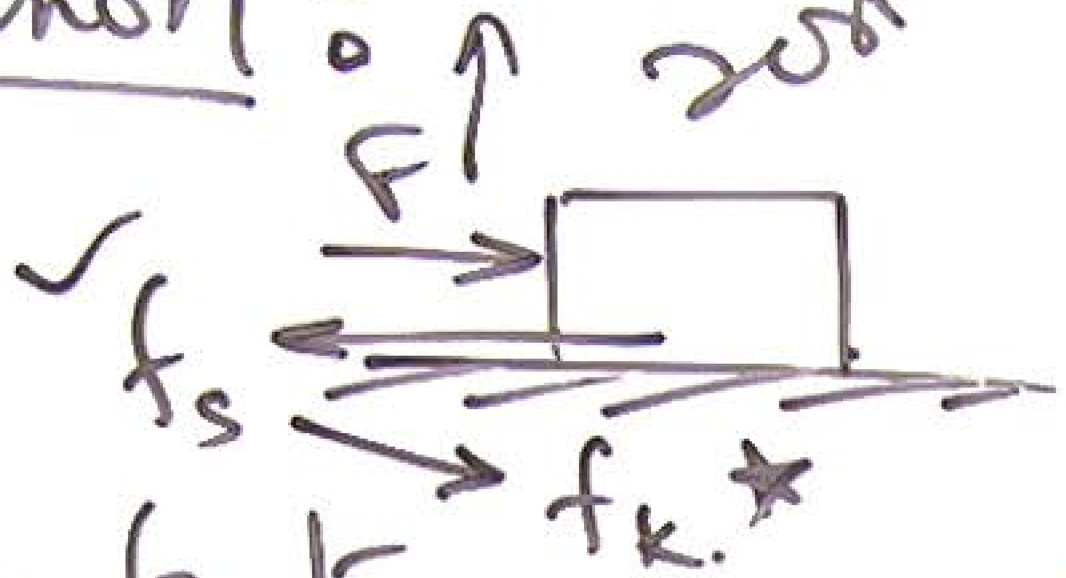
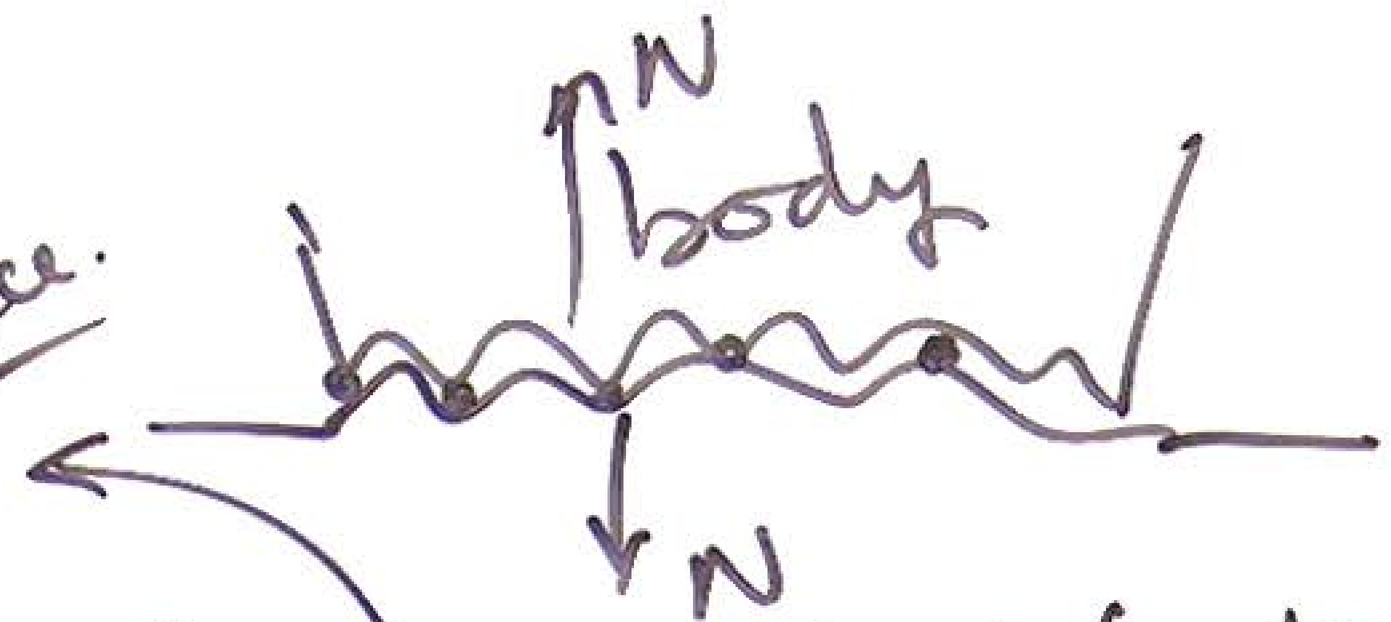
① Types of friction :



$f_s = F$.



- Viscous or Wet friction
- Rolling friction — due to deformation of materials during rolling
- Sliding friction — due to relative motion between two surfaces in contact or even by tendency of rel motion

Static friction

($f_{s\,max}$) Limiting friction

* Kinetic friction

f_k.

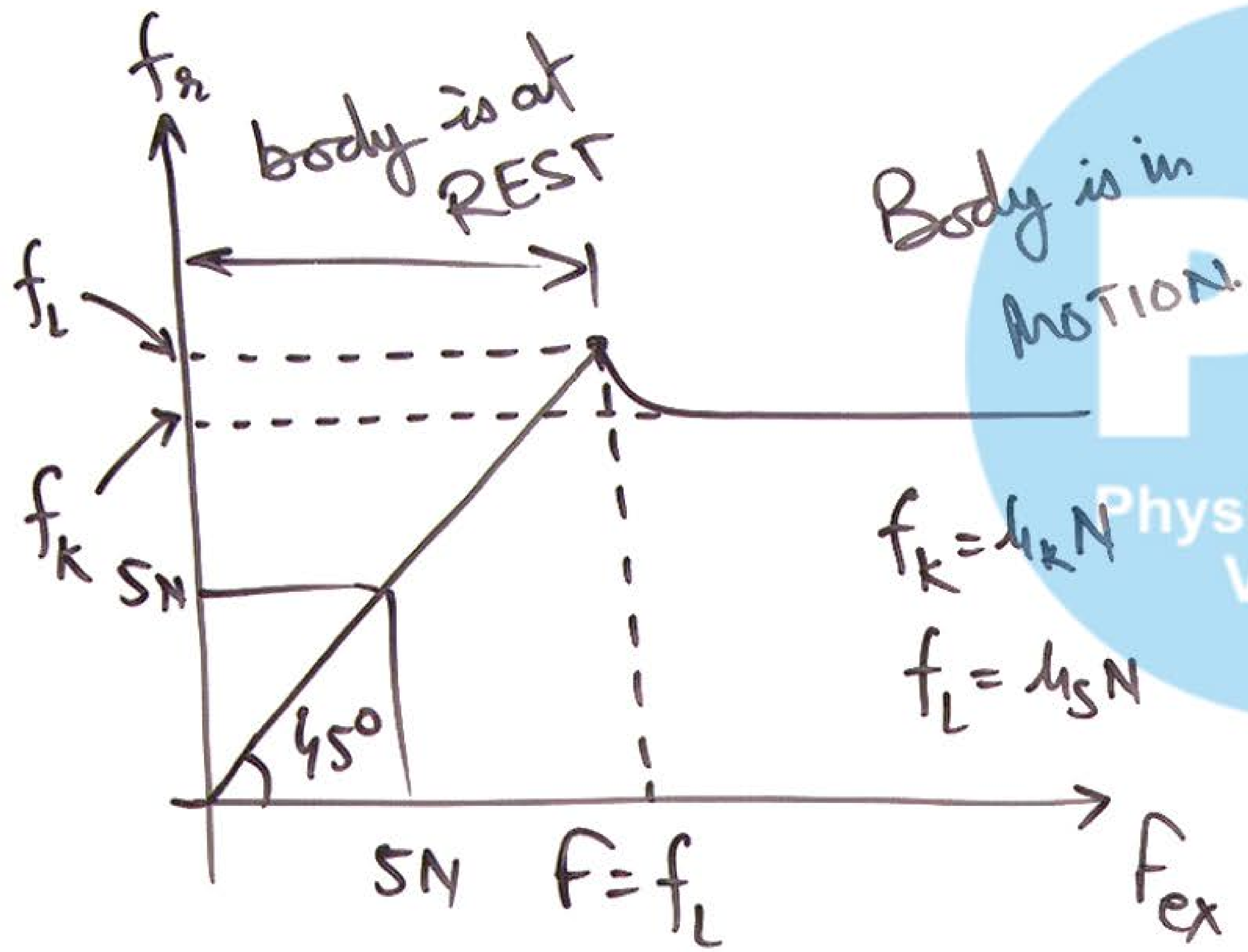
$(f_k < f_{s\,max})$

$\mu_k < \mu_s$

$f_k = \mu_k N$

$f_{s\,max} = \mu_s N$

(2) Variation in friction with external force:



NOTE: in Some Cases where $\mu_S = \mu_K = \mu$.

body will not start sliding at $F = \mu N$.

if $\mu_K < \mu_S$ (given values)

$\Rightarrow$ body starts sliding at $F = \mu_S N$.

(3) Pulling is easier than Pushing On a Surface :

to slide the block $T\cos\theta = \mu(mg - T\sin\theta)$

$f_{r_2} = \mu_k N_2$

$N_2 = mg - T\sin\theta$

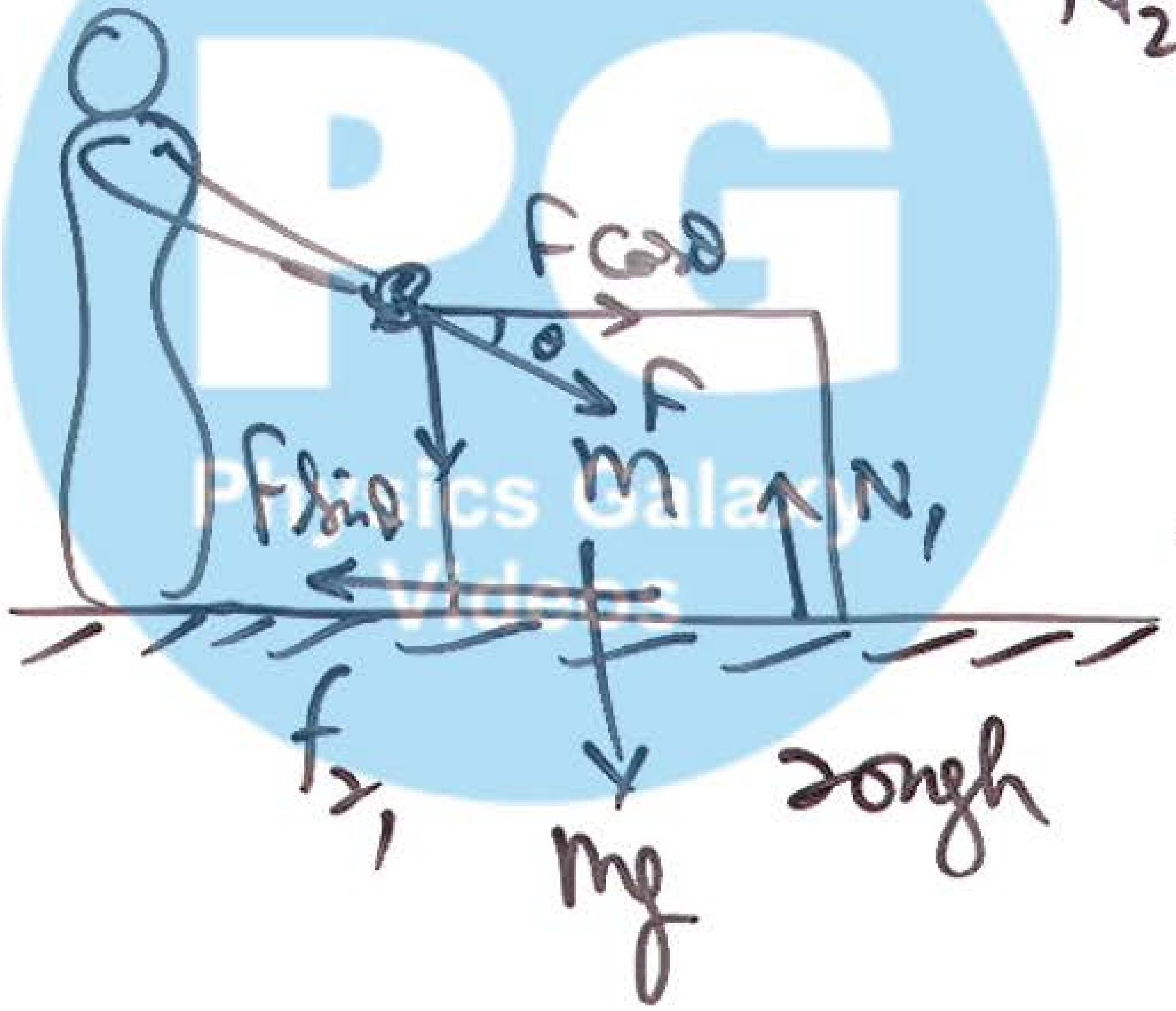
$T = f(\theta)$

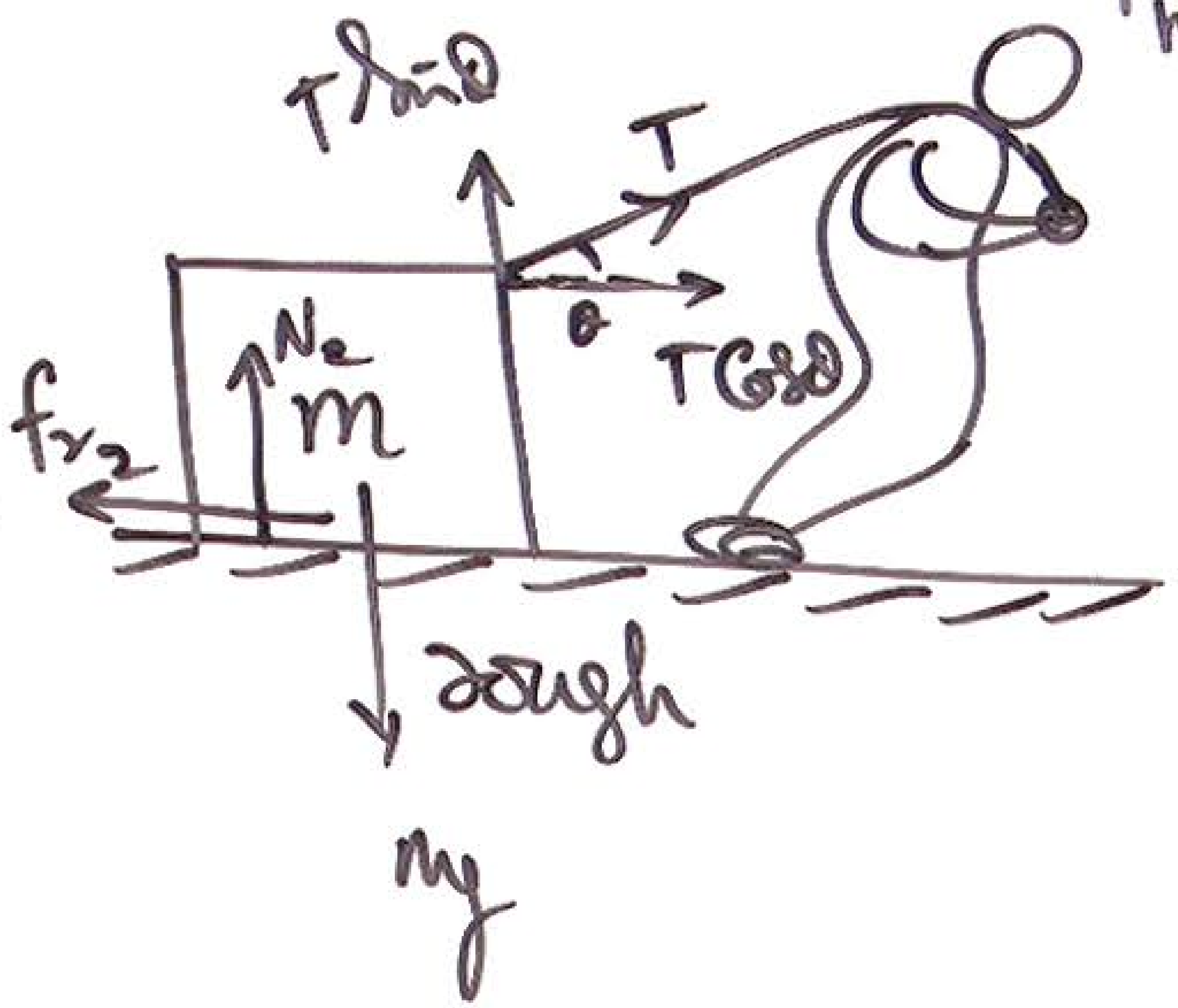
$\frac{dT}{d\theta} = 0$

$T_{min} = $ —

$N_1 = mg + F\sin\theta$

$f_{r_1} = \mu_k N_1$

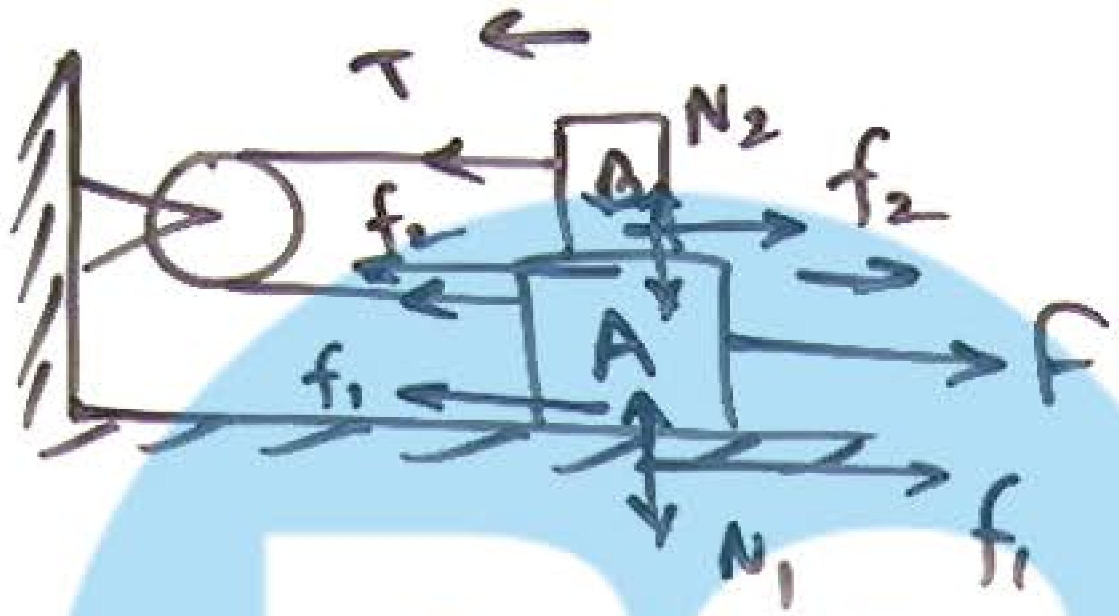


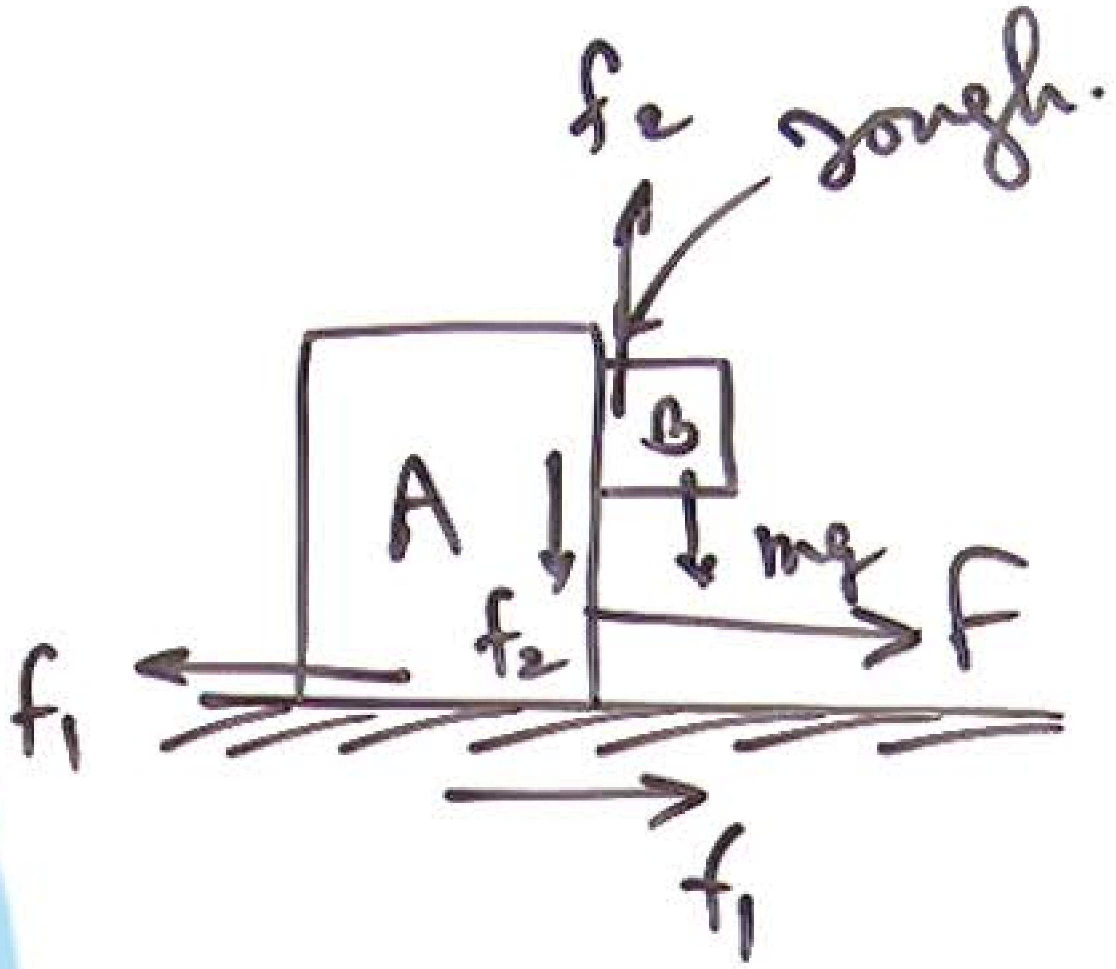


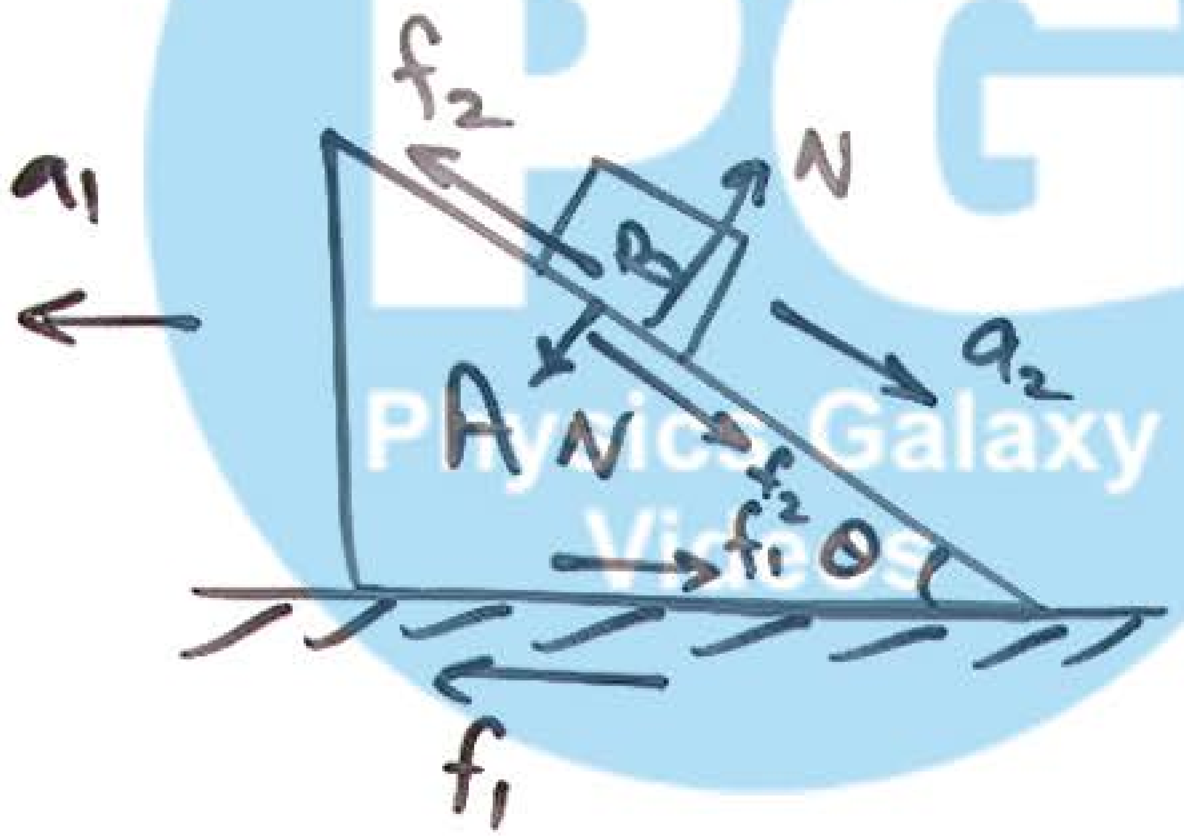
(4) friction in pulley & block cases:

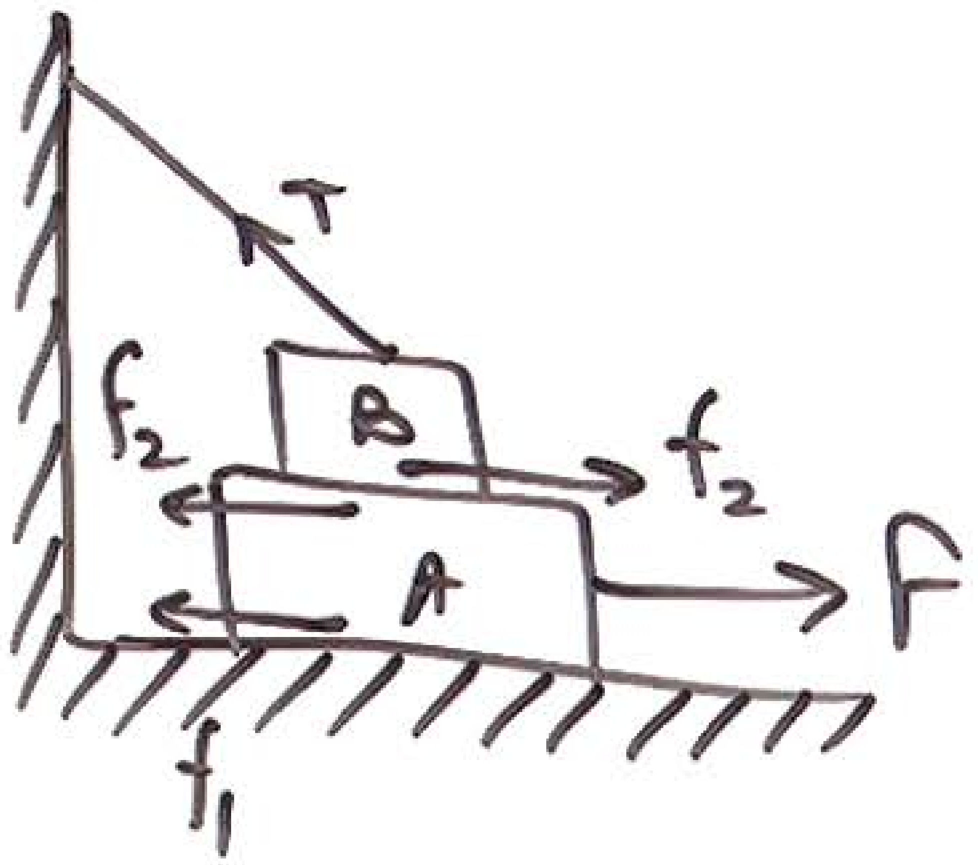
most imp is dir of friction

rest analysis of case is done by NLM concept.

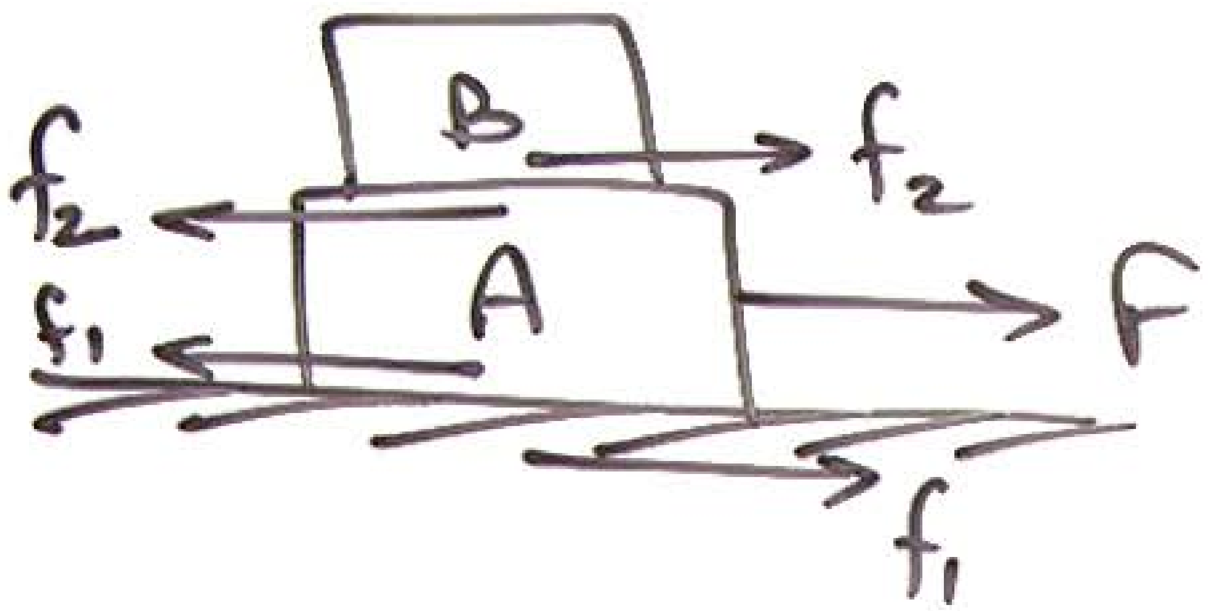








5 Block over Block:



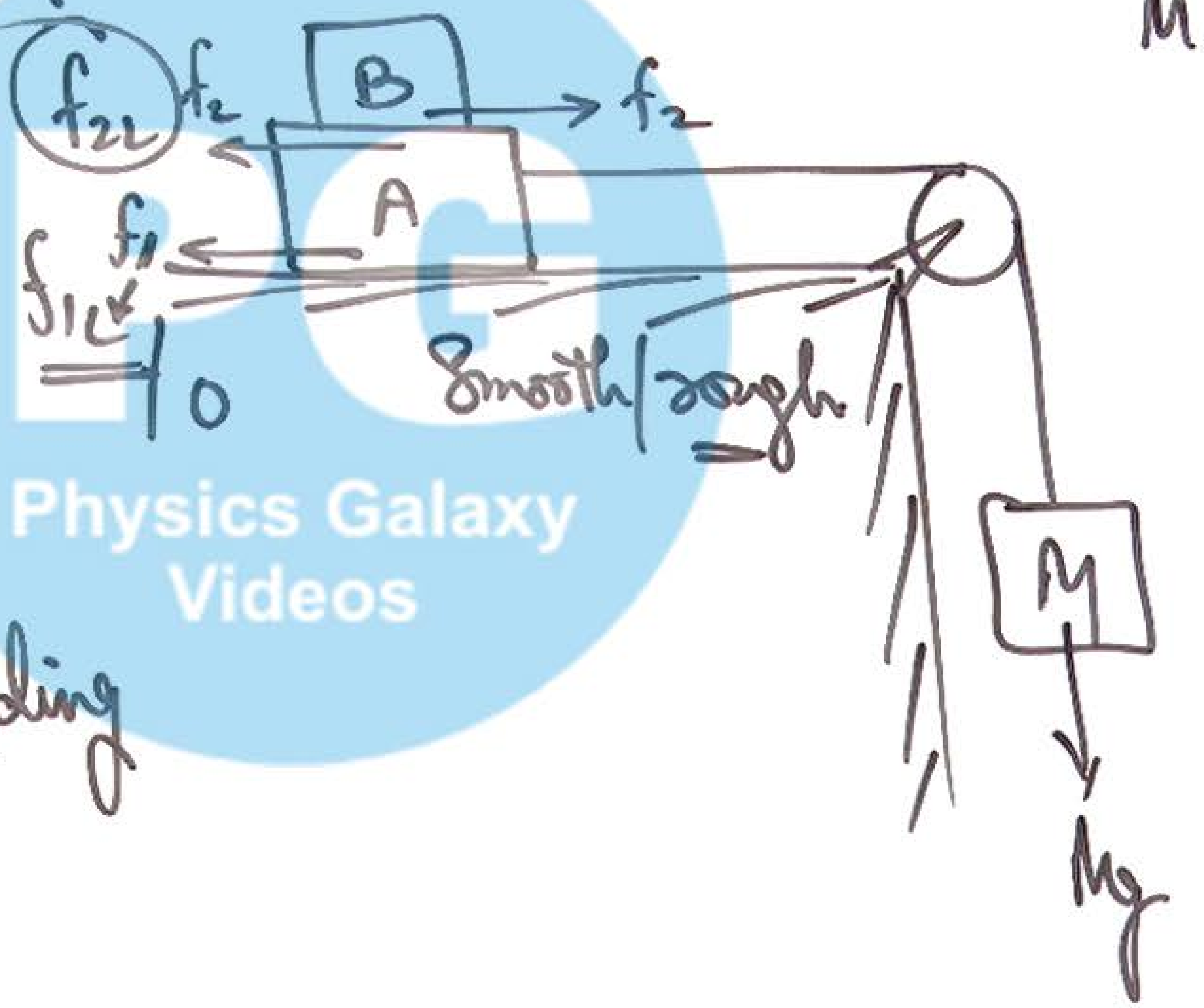
B can never slide unless A starts sliding

Eqⁿ of motion of B

$$f_2 = m_B \cdot a$$

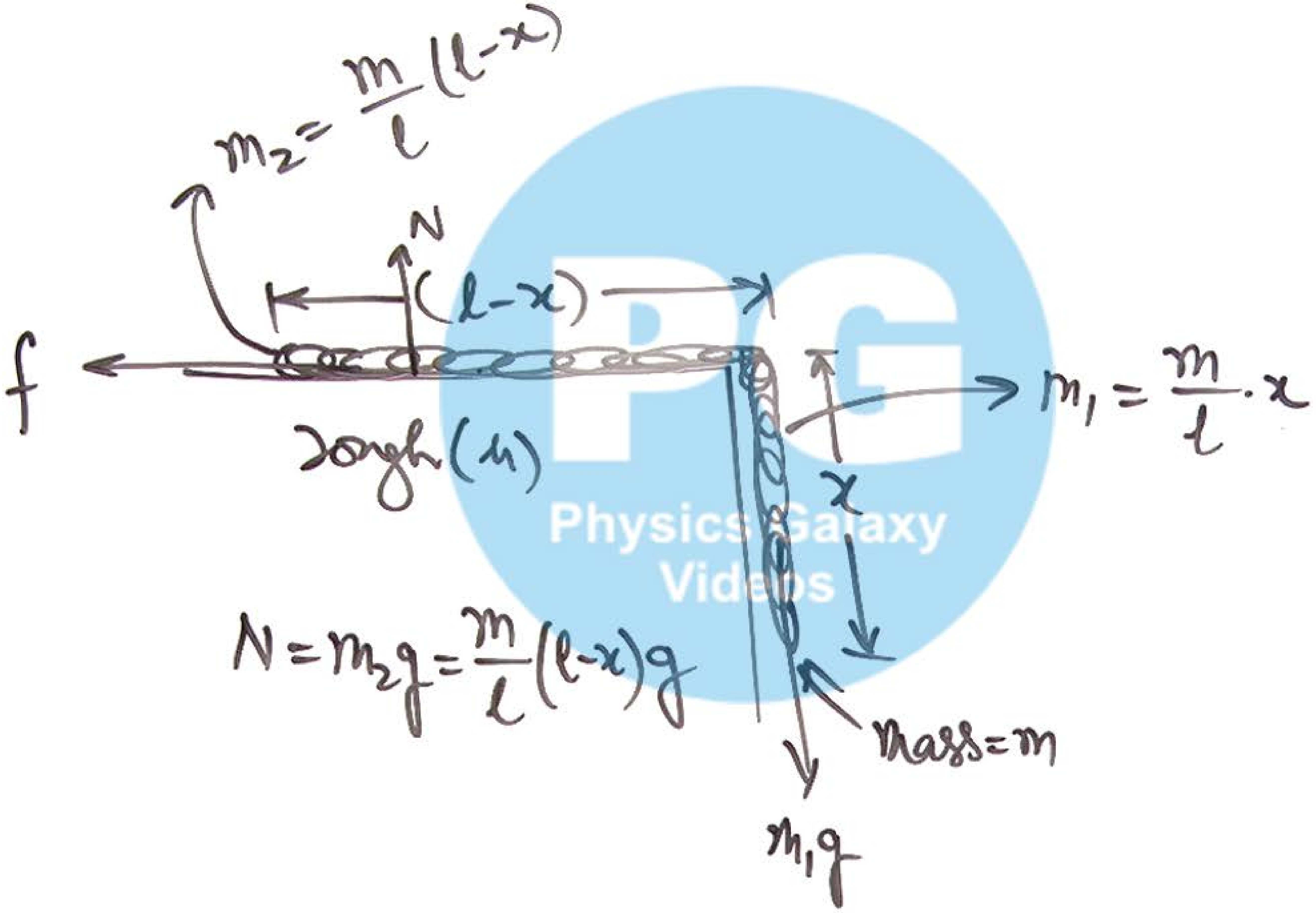
M = ✓

$$\rightarrow \quad a = \frac{Mg - f_{1L}}{M + m_A + m_B}$$



find M=? for which B starts sliding over A.

(6) Friction on a chain :



$$N = m_2 g = \frac{m}{l}(l-x)g$$

If chain is at rest

$$m_1 g = f_s.$$

If chain is sliding

$$f = \mu N$$

$$f = \frac{\mu m}{l}(l-x)g$$